

Devansh Garg

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Research Interests

Systems for large-scale machine learning: AI inference optimization, in-network computing, and datacenter networking — congestion control, RDMA, and programmable data planes.

Education

- IIT Hyderabad**, M.Tech. Computer Science & Engineering Jul 2025 – May 2028
- **Coursework:** Advanced Data Structures, Program Verification, Security in Intelligent Systems, Visual Computing, Topics in Networks CGPA: 9.2/10.0
- NIT Goa**, B.Tech. Computer Science & Engineering Nov 2021 – May 2025
- **Coursework:** Computer Networks, Computer Organization and Architecture, Theory of Computation CGPA: 8.9/10.0
- IIT Hyderabad**, Final-Year CSE Exchange Student Aug 2024 – May 2025
- **Coursework:** Advanced Topics in Cryptology, Network Security, Quantum Computing, Machine Learning

Publications

Detecting Manipulation to Table Rules in the Programmable Data Planes

Ranjitha K, Karuturi Havya Sree, **Devansh Garg**, Stavan Nilesch Christian, Dheekshitha Bheemanath, Rinku Shah, Praveen Tammana

IFIP Networking 2025, Limassol, Cyprus. pp. 438–446. [PDF]

XF-SISS: XOR-Based Fault Tolerant and Flexible Secret Image Sharing Scheme

Devansh Garg, B. R. Purushothama

IEEE OCIT 2025. pp. 81–86. [IEEE Xplore]

Manuscripts Under Review

Detecting Manipulation to Table Rules in the Programmable Data Planes

Ranjitha K, **Devansh Garg**, Rinku Shah, Praveen Tammana

IEEE Transactions on Network and Service Management (TNSM) — submitted August 2026, under review.

Extended version of the IFIP Networking 2025 paper above.

Research Experience

- Graduate Researcher**, IIT Hyderabad — advised by Dr. Praveen Tammana Jul 2025 – Present
- Software-systems-level optimization for AI inference, targeting the cost break-even point at which self-hosting open-weight models becomes cheaper than commercial API access.
 - Datacenter congestion control for high-performance networking, using DPDK and RDMA.
- In-Network Computing over Quantized Data**, IIT Hyderabad — course project Jul – Nov 2026
- Designing hardware support so that in-network computation can operate directly on quantized data, accelerating distributed training and inference.
- Demonstration System for P4Auth**, IIT Hyderabad 2025
- Built the live demonstration for P4Auth, showing three configurations side by side: baseline HULA load balancing, HULA under a compromised switch, and HULA protected by P4Auth.
 - Acknowledged in *Securing In-Network Traffic Control Systems with P4Auth* (IEEE/IFIP DSN 2025). [IEEE Xplore]

Programmable Data Plane Security (P4TVal), IIT Hyderabad

Aug 2024 – Apr 2025

- Implemented Rule Enforcement Verification (REV), the state-of-the-art baseline, in a BMv2–Mininet testbed using P4; simulated adversarial table-rule modification via the switch CLI.
- Measured attack-detection time against network size for both systems, establishing that P4TVal detects rule manipulation $10\times$ faster than REV and stays constant as hop count grows.
- Evaluated data-plane overhead on an Aurora 610 Intel Tofino switch (bf-sde 9.9.0, PTF): ingress-pipeline latency distribution over 5,000 packets, showing ~ 50 ns added per packet. [code]

Cybersecurity Research Intern, NIT Surathkal

Jun 2024 – Jul 2024

- Designed a flexible XOR-based secret image sharing scheme enabling partial recovery with perfect secrecy; published as XF-SISS at IEEE OCIT 2025.
- Evaluated PSNR benchmarks over large image datasets using Python, OpenCV, and NumPy.

Projects

BoSy — Reactive Synthesis Tool Extensions, IIT Hyderabad — course project, team of 3

Jan – May 2026

- Added AIGER circuit output and integrated the Pedant DQBF solver into BoSy, an open-source bounded-synthesis tool written in Swift.
- Authored the technical documentation for the extended toolchain, covering solver backends, bound-search strategies, and a guide to the codebase. [code]

Teaching

Teaching Assistant, Network Security (CS6903) — IIT Hyderabad

Jan – May 2026

- Graduate-level course, with Dr. Praveen Tammana and Dr. Saurabh Kumar.
- Designed capture-the-flag problems, quizzes, and assignments; conducted vivas, graded submissions, and invigilated examinations.

Teaching Assistant, Computer Networks (CS3530) — IIT Hyderabad

Jul – Nov 2025

- Undergraduate course, with Dr. Praveen Tammana.
- Prepared quizzes and assignments, graded submissions, and invigilated examinations.

Industry Experience

GenAI Intern, Mondee Tech

May 2025 – Jul 2025

- Built a multi-agent system with live voice interaction over WebSockets using Gemini Live; integrated LLMs, VLMs, STT/TTS, and OCR models into agent workflows.
- Engineered FastAPI backends and deployed them to production.

Technical Skills

Languages: C++, C, Python, P4, Bash, SQL

Networking & Systems: Intel Tofino, BMv2, Mininet, P4Runtime, PTF, DPDK, RDMA, Linux, gRPC, Git

ML / AI: vLLM, LLMs, VLMs, OCR, agent workflows, LangGraph, Hugging Face

Other: FastAPI, WebSockets, PostgreSQL, MySQL, OpenCV, NumPy, Google Cloud